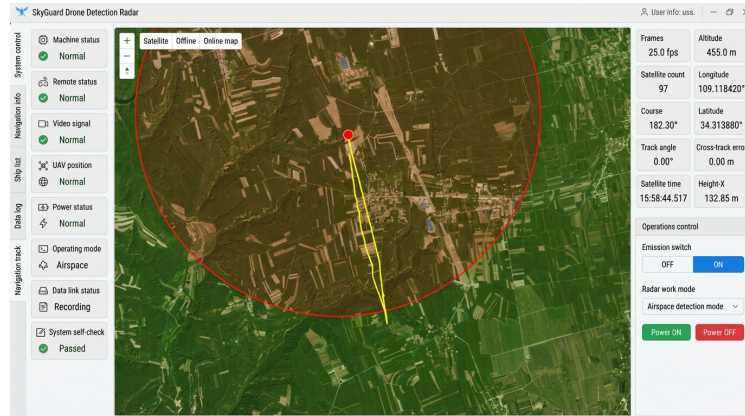


Sky-Guard Console

Digital Intelligent Flight Decision Support System



Background

As the low-altitude economy accelerates and public interest in recreational aviation grows, flight camps — serving general aviation, paragliding, drone testing, powered hang-gliders, hot-air balloons, and other light aircraft — are transitioning from experience-based management to data-driven, intelligent operations. Weather conditions such as wind speed, temperature, humidity, barometric pressure, cloud ceiling, and precipitation have a decisive impact on the safety and performance of light aircraft. The Sky-Guard Console bridges the gap between raw meteorological data and actionable flight decisions.

Core Capabilities

- **Real-Time Weather Assessment:** Continuously evaluates whether current conditions meet safe takeoff and landing thresholds for each aircraft type.
- **Flight Simulation:** Models flight attitude, climb rate, glide ratio, and control difficulty under varying meteorological scenarios.
- **Optimal Flight Planning:** Recommends best takeoff/landing direction, timing, climb profiles, holding patterns, and return-to-home waypoints.
- **Ideal Flight Window Prediction:** Forecasts time-slot-based operating windows to optimise camp scheduling and customer booking.
- **Automated Risk Alerts:** Generates flight-risk warnings and operational recommendations to support instructors and pilots in real-time decision making.
- **Personalised Flight Advice:** Tailors recommendations based on site terrain, user profile (age, activity intensity), and spatiotemporal context.

Personalised Camp Features

- Individual flight decision recommendations per visitor
- 3-day advance safety risk forecast for the camp
- Optimal flight period suggestions
- Visual flight-path preview and simulation

- Locally-tuned AI decision model that improves over time
- Emergency / sudden-change weather alerts

Solution Architecture

The system fuses real-time meteorological data ingestion, an aircraft performance parameter library, aerodynamic models, and AI-assisted decision algorithms into a unified platform. Data sources include gridded weather model outputs (1–2 km resolution), on-site sensor calibration, and historical operational records.

Service Tiers

Feature	Standard	Professional	Enterprise
Core Output	Index monitoring & review	Index-driven decision loop	Full custom pipeline
Forecast Horizon	24 h rolling	48 h rolling	72–120 h trend
Monitored Points	5 custom locations	10 custom locations	10+ (expandable)
AI Service	—	Multi-index analytics	Self-learning model
Alert Channel	Web dashboard	Real-time push notifications	Multi-channel + audit trail
Deliverables	Flight data logging, 24 h forecast, 5-point report, web portal	Daily report + video briefing, 48 h forecast, 10-point report, push notifications	Custom decision reports, flight behaviour logging, 72–120 h forecast, 10–15 point report, multi-system push

Compliance & Risk Boundaries

- The service provides indices and advisory actions only; it does not replace any certified environmental data product.
- Indices serve as decision support, not the sole basis for safety or regulatory compliance. Operators must retain final authority and on-site verification procedures.
- When historical data is insufficient or significantly biased, model uncertainty will increase. The system flags such conditions and supports fallback to rule-based / conservative strategies.

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The information contained in this document is provided for reference purposes only. Specifications are subject to change without prior notice. Actual performance may vary depending on operating environment, deployment configuration, and integration conditions.